

Higher-order clustering for 3D point clouds

Exact density hierarchies and geometric tokens for 3D learning

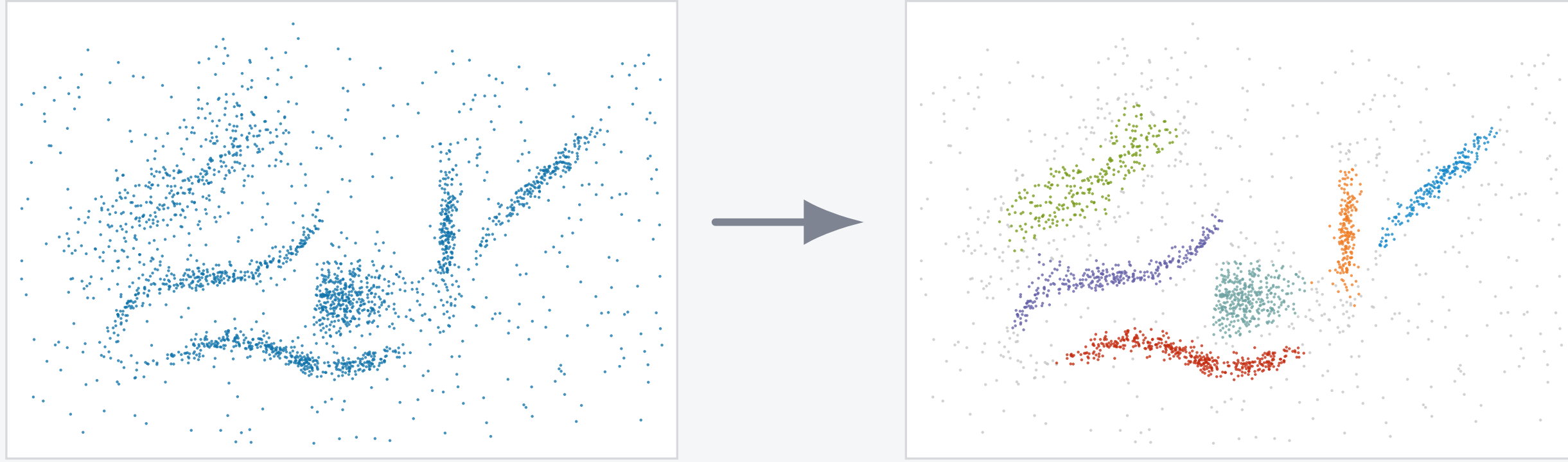
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Clustering

Clustering groups similar points into *clusters*. Here, similarity is measured by Euclidean distance.

Unlabelled points

HDBSCAN [3]



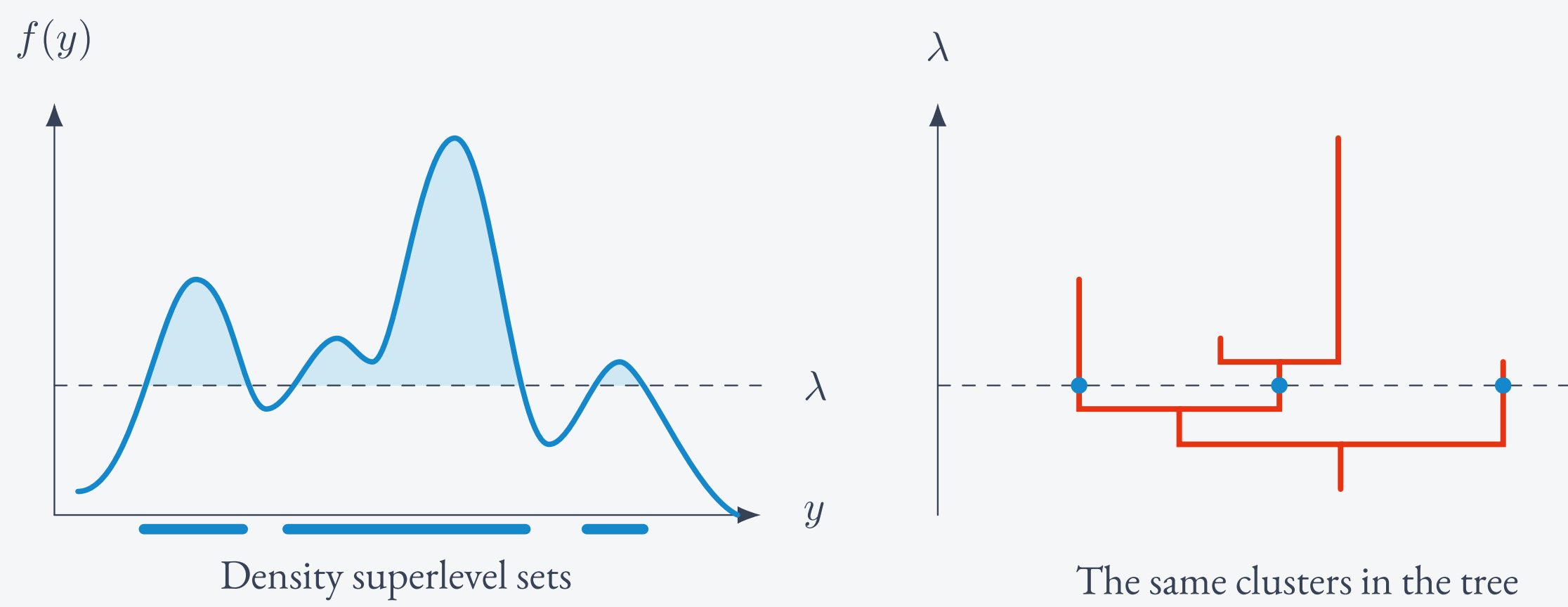
Clusters in colour; unassigned points in grey [3].

HARTIGAN's model and hierarchy

For a density f , clusters are the **connected components** of its superlevel sets [1].

$$E_\lambda = \{y \in \mathbb{R}^d : f(y) \geq \lambda\}$$

Lowering λ makes components grow and merge: a **hierarchy**.



We seek to **estimate the unknown density f** . A natural, non-parametric choice is the K -nearest-neighbour estimator [2]:

$$\hat{f} \leftarrow \hat{f}_{K\text{-NN}}$$

$$\hat{f}_{K\text{-NN}}(y) = \frac{K}{n\omega_d r_K(y)^d}$$

$r_K(y)$: distance to the K -th nearest sample point.

n : number of points; ω_d : unit-ball volume.

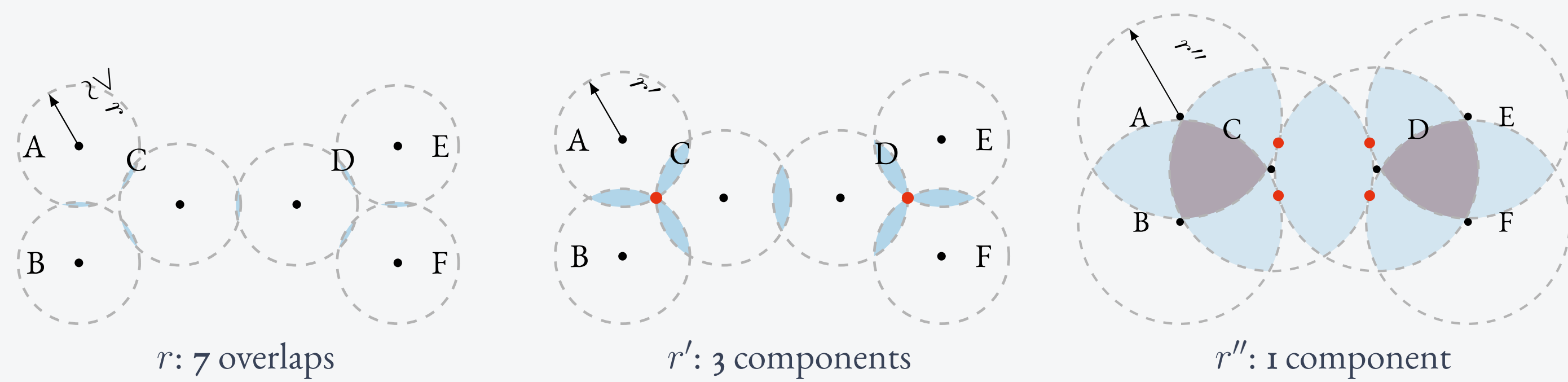
Change of variable: density level to neighbour radius.

$$\hat{f}_{K\text{-NN}}(y) \geq \lambda \iff r_K(y) \leq (K/(n\omega_d\lambda))^{1/d} =: r(\lambda)$$

Same six points, different hierarchies

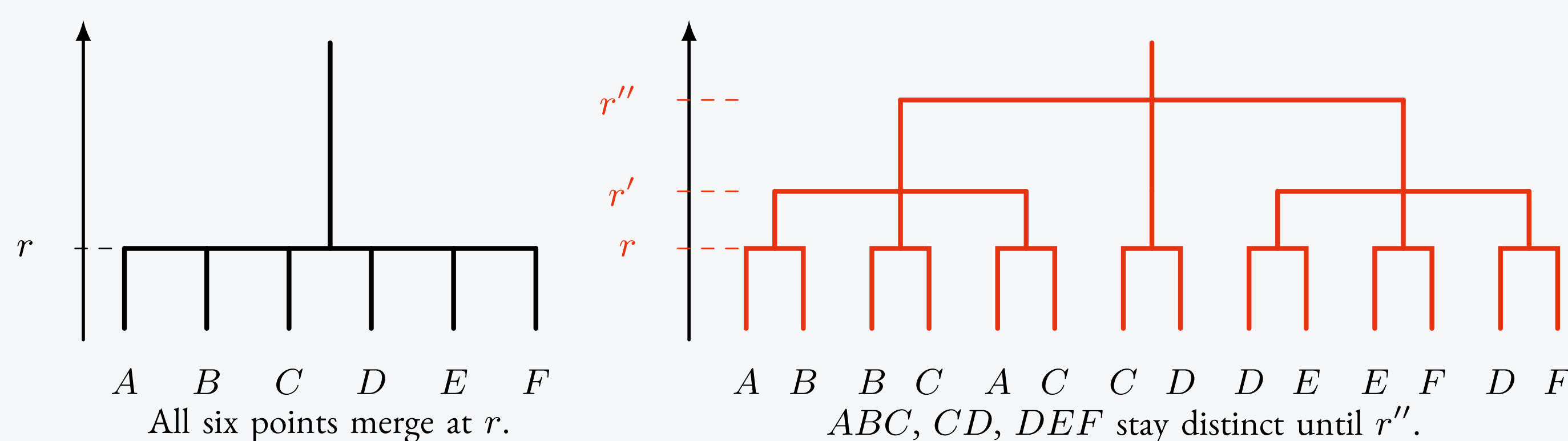
For $K = 2$, a density superlevel set is the region covered by **at least two balls**:

$$\{\hat{f}_{2\text{-NN}} \geq \lambda\} = \bigcup_{i < j} (B(x_i, r(\lambda)) \cap B(x_j, r(\lambda)))$$



DBSCAN / Robust SL

2-NN HARTIGAN



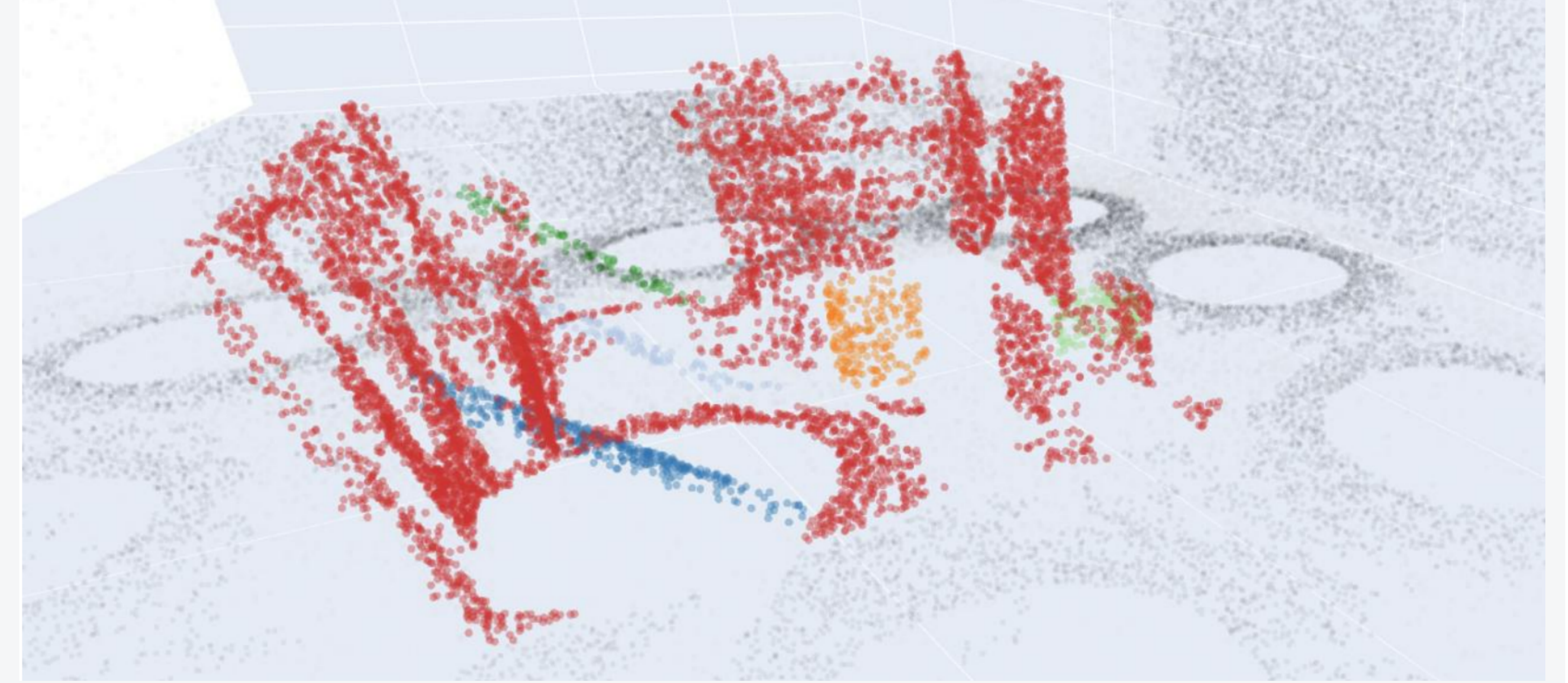
HGP-Clusterer connects edges through triangles to recover the exact hierarchy [4,5].

References

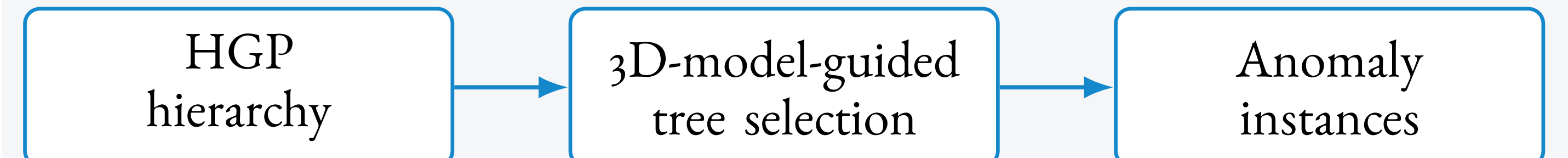
- [1] J. A. Hartigan. *Consistency of single linkage for high-density clusters*. JASA, 76:388–394, 1981.
- [2] G. Biau and L. Devroye. *Lectures on the Nearest Neighbor Method*. Springer, 2015.
- [3] R. J. G. B. Campello, D. Moulavi and J. Sander. *Density-Based Clustering Based on Hierarchical Density Estimates*. PAKDD, pp. 160–172, 2013.

LiDAR: detect unexpected objects

Objects present in a scan, absent from a reference 3D model. With Marie ASPRO (Inria Startup Studio).



Synthetic benchmark: 2 million points. Red: model-consistent; other colours: anomalies; grey: background. © Naval Group [6].



Geometric priors select branches. No training required.

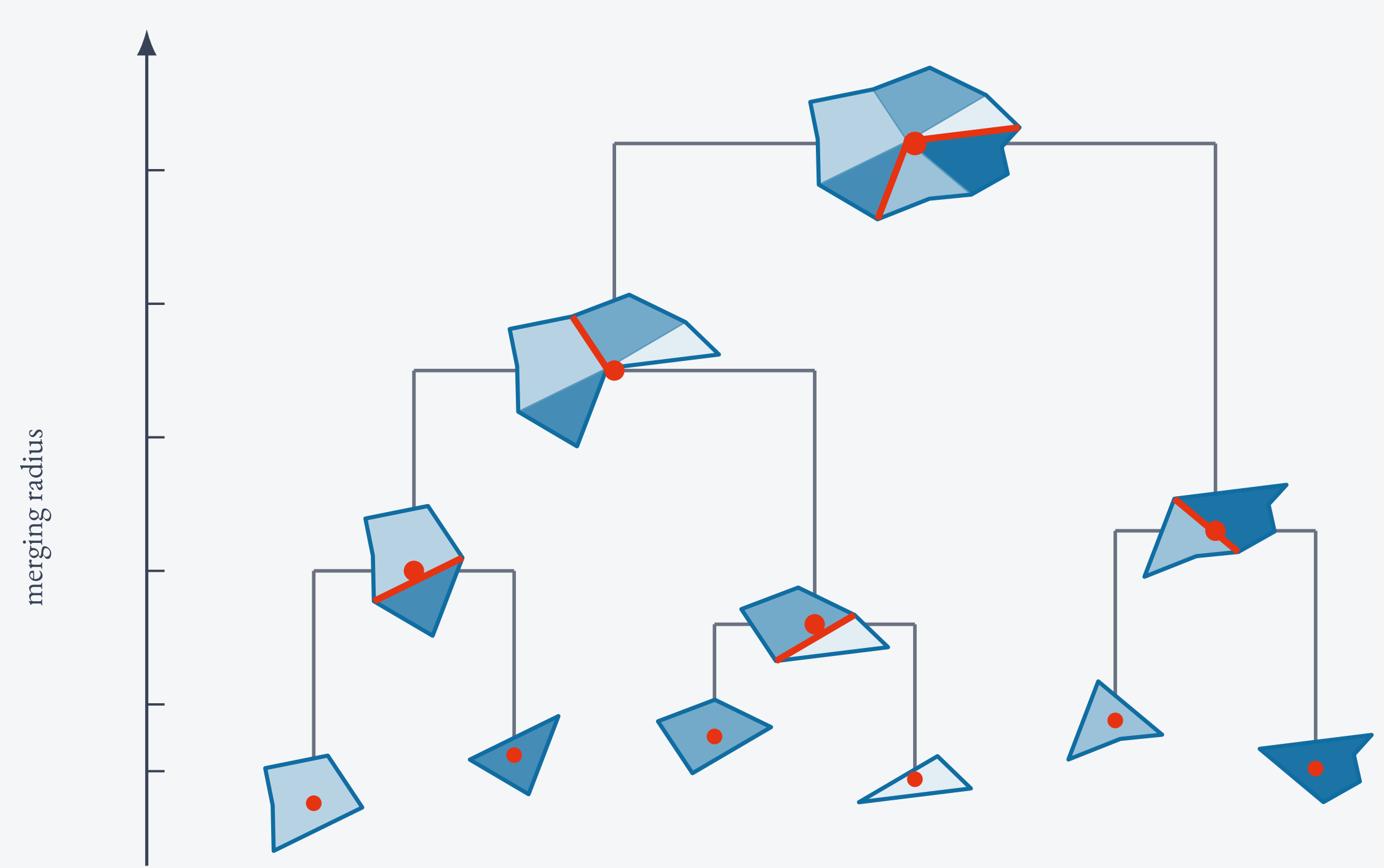
Perspective: geometry-guided 3D learning

The same object, different sampling.

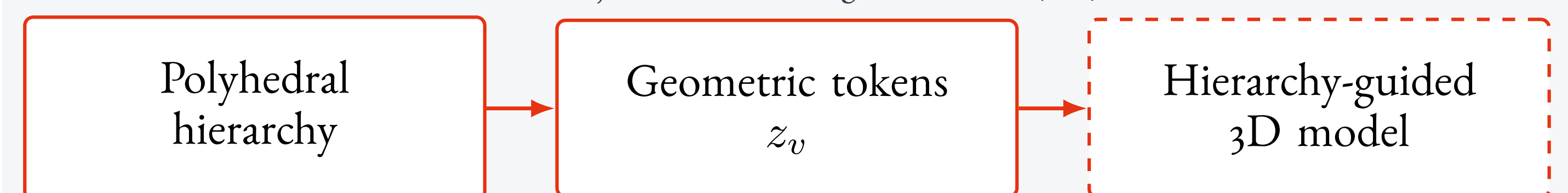


Range, scan pattern and occlusion change the point cloud.

Polyhedra as tokens; hierarchy as multiscale context.



Parents join children along shared faces (red).



Research hypothesis: use polyhedral tokens and merging scales to guide a 3D foundation model.

Inria Startup Studio: Percolia



Louis HAUSEUX & Alban HAUSEUX
Entering Inria Startup Studio.
Geometry-aware tools for 3D perception.



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